

12 General Renormalization Theory

the amplitude corresponding to any graph is that power-counting should give $D < 0$ not only for the complete multiple integral for the whole amplitude, but also for any subintegration defined by holding any one or more linear combinations of the loop momenta fixed. (The graphs shown in Figures 12.1(b) and 12.1(c) fail this test because $D \geq 0$ for the subintegrations in which only the momenta for the loops within the dotted squares are integrated.) We will not repeat the rather long proof here, because it is well treated in earlier books [3], and in any case the method of proof has little to do with how we actually do calculations. The next section will describe how this requirement is fulfilled.

12.2 Cancellation of Divergences

Consider a Feynman diagram, or part of a Feynman diagram, with positive superficial degree of divergence, $D \geq 0$. The part of the momentum space integral where all internal momenta go to infinity together will then diverge, like $\int^\infty k^{D-1} dk$. If we differentiate $D+1$ times with respect to any external momentum, we lower the net number of momentum factors in the integrand by $D+1$,¹ and hence render this part of the momentum space integral convergent. There may still be divergences arising from subgraphs, like those in Figures 12.1(b) and 12.1(c); for the moment we will ignore this possibility, returning to it later in this section. Since differentiation $D+1$ times renders the integral finite, it follows that the contribution of such a graph or subgraph can be written as a polynomial of order D in external momenta, with divergent coefficients, plus a finite remainder.

To see how this works without irrelevant complications, consider the logarithmically divergent one-dimensional integral

$$\mathcal{J}(q) = \int_0^\infty \frac{dk}{k+q}$$

with $D = 1 - 1 = 0$. Differentiating once gives

$$\mathcal{J}'(q) = - \int_0^\infty \frac{dk}{(k+q)^2} = -\frac{1}{q},$$

so

$$\mathcal{J}(q) = -\ln q + c.$$

The constant c is obviously divergent, but the rest of the integral is perfectly finite. In exactly the same way, we can evaluate the $D = 1$ integral

$$\int_0^\infty \frac{k dk}{k+q} = a + bq + q \ln q$$

with divergent constants a and b .

¹For instance, if an internal scalar field line carries a momentum $k+p$, where p is a linear combination of external four-momenta and k is a four-momentum variable of integration, then the derivative of the propagator $[(k+p)^2 + m^2]^{-1}$ with respect to p^μ gives $-2(k_\mu + p_\mu)[(k+p)^2 + m^2]^{-2}$, which goes as k^{-3} rather than k^{-2} for $k \rightarrow \infty$.

Now, a polynomial term in external momenta is just what would be produced by adding suitable terms to the Lagrangian: if a graph with E_f external lines of type f has degree of divergence $D \geq 0$, then the ultraviolet divergent polynomial is the same as would be produced by adding various interactions i with $n_{if} = E_f$ fields of type f and $d_i \leq D$ derivatives. If there already are such interactions in the Lagrangian, then the ultraviolet divergences simply add corrections to the coupling constants of these interactions. Hence these infinities can be cancelled by including suitable infinite terms in these coupling constants. All that we ever measure is the sum of the bare coupling constant and the corresponding coefficient from one of the divergent polynomials, so if we demand that the sum equals the (presumably finite) measured value, then the bare coupling must automatically contain an infinity that cancels the infinity from the divergent integral over internal momenta. (One qualification: where the divergence occurs in a graph or subgraph with just two external lines, which appears as a radiative correction to a particle propagator, we must demand not that some effective coupling constant equals its measured value, but rather that the complete propagator has a pole at the same position and with the same residue as for free particles.) In this way, all infinities are absorbed into a redefinition of couplings constants, masses, and fields.

For this renormalization program to work, it is essential that the Lagrangian include *all* interactions that correspond to the ultraviolet divergent parts of Feynman amplitudes. (There are exceptions to this rule in supersymmetric theories [4].) The interactions in the Lagrangian are, of course, limited by various symmetry principles, such as Lorentz invariance, gauge invariance, etc., but these constrain the ultraviolet divergences in the same way. (It takes some work to prove that non-Abelian gauge symmetries constrain infinities in the same way that they constrain interactions. This will be shown in Volume II.) In the general case, there are no other limitations on the ultraviolet divergences, so *the Lagrangian must include every possible term consistent with symmetry principles.*

But there is an important class of theories with only a finite number of interactions, where the renormalization program also works. These are the so-called renormalizable theories, whose interactions all have $\Delta_i \geq 0$. Equation (12.1.8) then gives

$$D \leq 4 - \sum_f E_f (s_f + 1),$$

so divergent polynomials arise in only a limited number of Feynman graphs or subgraphs: those with few enough external lines so that $D \geq 0$. The contribution of such divergent polynomials is just the same as would be produced by replacing the divergent graph or subgraph with a single vertex arising from a term in the Lagrangian with E_f fields of type f and $0, 1, \dots, D$ derivatives. But, comparing with Eq. (12.1.9), we see that *these are precisely the same as the interactions that satisfy the renormalizability requirement $\Delta_i \geq 0$, or in other words,*

$$0 \leq d_i \leq 4 - \sum_f n_{if} (s_f + 1).$$

In order for all infinities to cancel in a renormalizable theory, it is usually necessary that *all* renormalizable interactions that are allowed by symmetries must actually appear in the

Lagrangian.² For instance, if there is a scalar (or pseudoscalar) field ϕ and fermion field ψ with interactions $\bar{\psi}\psi\phi$ (or $\bar{\psi}\gamma_5\psi\phi$) then we cannot exclude an interaction ϕ^4 ; otherwise there would be no counterterm to cancel the logarithmic divergence arising from fermion loops with four attached scalar or pseudoscalar lines.

Let's see in more detail how the cancellation of infinities works in the simplest version of quantum electrodynamics. Equation (12.1.11) shows that the only graphs or subgraphs that could possibly yield divergent integrals are the following:

$$E_e = 2, E_\gamma = 1$$

This is the electron-photon vertex $\Gamma_\mu^{*(\ell)}(p', p)$. (The superscript ℓ indicates that this includes only contributions from graphs with loops.) It has $D = 0$, so its divergent part is momentum-independent. Lorentz invariance then only allows this divergent constant to be proportional to γ_μ , so

$$\Gamma_\mu^{*(\ell)} = L\gamma_\mu + \Gamma_\mu^{(f)}, \quad (12.2.1)$$

with L a logarithmically divergent constant, and $\Gamma_\mu^{(f)}$ finite. This does not uniquely define the constant L , since we can always move a finite term $\delta L\gamma_\mu$ from $\Gamma_\mu^{(f)}$ to $L\gamma_\mu$. To complete the definition, we may note that as shown in Section 9.7, the mass-shell matrix element of $\Gamma_\mu^*(p, p)$ and hence of $\Gamma_\mu^{(f)}(p, p)$ between mass-shell Dirac spinors is proportional to the same matrix element of γ^μ , so we may define L by the prescription that

$$\bar{u}(p, \sigma')\Gamma_\mu^{(f)}(p, p)u(p, \sigma) = 0 \quad (12.2.2)$$

for $p^2 + m_e^2 = 0$.

$$E_e = 2, E_\gamma = 0$$

This is the electron self-energy insertion $\Sigma^*(p)$. It has $D = 1$, so its divergent part is linear in the momentum p^μ carried by the incoming and outgoing fermion. Lorentz invariance (including parity conservation) will only allow it to be a function of $\not{p}$, so we may write the loop contribution as

$$\Sigma^{*(\ell)}(p) = A - (i\not{p} + m)B + \Sigma^{(f)}(\not{p}), \quad (12.2.3)$$

where A and B are divergent constants, and $\Sigma^{(f)}$ is finite. Again, this does not uniquely define the constants A and B , because we can always shift $\Sigma^{(f)}$ by a finite first-order polynomial in $\not{p}$. We will define A and B by the prescription that

$$\Sigma^{(f)} = \frac{\partial \Sigma^{(f)}}{\partial \not{p}} = 0 \quad \text{for} \quad i\not{p} = -m. \quad (12.2.4)$$

Actually, B is not a new divergent constant. As long as we use a regularization procedure that respects current conservation, Γ_μ and Σ will be related by the Ward identity (10.4.27)

$$\Gamma^\mu(p, p) = \gamma^\mu + i \frac{\partial}{\partial p_\mu} \Sigma(p)$$

²In addition, interactions and mass terms that are not allowed by global symmetries may appear in the Lagrangian, as long as they are superrenormalizable, that is, with $\Delta_i > 0$. This is because the presence of a superrenormalizable coupling lowers the degree of divergence, so that the symmetry breaking does not affect those divergences that are cancelled by the strictly renormalizable couplings with $\Delta_i = 0$. Note that it is the *bare* strictly renormalizable couplings that would exhibit the symmetry; renormalized couplings that are defined in terms of mass-shell matrix elements generally show the effect of symmetry breaking.

and therefore

$$L\gamma_\mu + \Gamma_\mu^{(f)}(p, p) = B\gamma_\mu + i\frac{\partial\Sigma^{(f)}(p)}{\partial p^\mu}. \quad (12.2.5)$$

Taking the matrix element of this equation between $\bar{u}(p, \sigma')$ and $u(p, \sigma)$ and using Eqs. (12.2.2) and (12.2.4), we find

$$L = B. \quad (12.2.6)$$

$E_\gamma = 2, E_e = 0$

This is the photon self-energy insertion $\Pi_{\mu\nu}^*(q)$. It has $D = 2$, so its divergent part is a second-order polynomial in q . Lorentz invariance only allows $\Pi_{\mu\nu}^*$ to take the form of a linear combination of $\eta_{\mu\nu}$ and $q_\mu q_\nu$ with coefficients depending only on q^2 , so the loop contributions take the form

$$\Pi_{\mu\nu}^{*(\ell)}(q) = C_1\eta_{\mu\nu} + C_2\eta_{\mu\nu}q^2 + C_3q_\mu q_\nu + \text{finite terms},$$

where C_1, C_2 , and C_3 are divergent constants. As long as we use a regularization technique that respects current conservation, we must have

$$q^\mu \Pi_{\mu\nu}^{*(\ell)}(q) = 0.$$

The same must then be true for the divergent terms, so $C_1q_\nu + (C_2 + C_3)q^2q_\nu$ must be finite for all q . It follows that C_1 and $C_2 + C_3$ must be finite, and can therefore be lumped into the finite part of $\Pi_{\mu\nu}^{*(\ell)}(q)$. Thus

$$\Pi_{\mu\nu}^{*(\ell)}(q) = (\eta_{\mu\nu}q^2 - q_\mu q_\nu)(C + \pi(q^2)), \quad (12.2.7)$$

where $\pi(q^2)$ is finite and C is the sole remaining divergence in $\Pi_{\mu\nu}$. To pin down the definition of C , we may move any finite constant $\pi(0)$ into C , so that

$$\pi(0) = 0. \quad (12.2.8)$$

$E_\gamma = 4, E_e = 0$

This is the amplitude $M_{\mu\nu\rho\sigma}$ for scattering of light by light. It has $D = 0$, so using Lorentz invariance and Bose statistics, it may be written (there is no non-loop contribution)

$$M_{\mu\nu\rho\sigma} = K(\eta_{\mu\nu}\eta_{\rho\sigma} + \eta_{\mu\rho}\eta_{\nu\sigma} + \eta_{\mu\sigma}\eta_{\nu\rho}) + \text{finite terms}$$

with K a potentially divergent constant. However, current conservation gives

$$q^\mu M_{\mu\nu\rho\sigma} = 0$$

and so $K(q_\nu\eta_{\rho\sigma} + q_\rho\eta_{\nu\sigma} + q_\sigma\eta_{\nu\rho})$ is finite. In order for this to be true for $q \neq 0$, K must itself be finite. This is a nice example of the role of symmetry principles in the renormalization program: if K had turned out to be infinite it could not be removed by renormalization of the coupling constant for an interaction $(A_\mu A^\mu)^2$, because no such interaction is allowed by gauge invariance, but K is finite because of current conservation conditions that are imposed by gauge invariance.

$E_\gamma = 1, E_e = 0$ **and** $E_e = 1, E_\gamma = 0, 1, 2$

These have $D = 3$ and $D = \frac{5}{2}, \frac{3}{2}$, and $\frac{1}{2}$, respectively, but Lorentz invariance makes all such graphs vanish.

$E_\gamma = 3, E_e = 0$

This has $D = 1$, but vanishes because of charge-conjugation invariance.

The reader will perhaps have noticed that the independent divergent constants A, B, C are in one-to-one correspondence with the independent parameters Z_2, Z_3 , and δm in the counterterm part (11.1.9) of the Lagrangian for quantum electrodynamics. These counterterms make a direct contribution $Z_2\delta m - (Z_2 - 1)(i\not{p} + m)$ to $\Sigma^*(p)$. The requirement that the position and residue of the one-particle pole be the same as in the free-field propagator means we must choose Z_2 and δm so that the total $\Sigma^*(p)$ satisfies Eq. (12.2.4), i.e.,

$$Z_2\delta m = -A, \quad (12.2.9)$$

$$Z_2 - 1 = -B, \quad (12.2.10)$$

so that the complete electron self-energy insertion is just the finite function $\Sigma^{(f)}(p)$:

$$\Sigma(p) = \Sigma^{(f)}(p). \quad (12.2.11)$$

Also, $\mathcal{L}_2$ makes a direct contribution to Γ_μ equal to $(Z_2 - 1)\gamma_\mu$. Using Eq. (12.2.6), we see that the full vertex is

$$\Gamma_\mu = \gamma_\mu + (Z_2 - 1)\gamma_\mu + \Gamma_\mu^{*(\ell)} = \gamma_\mu + \Gamma_\mu^{(f)}. \quad (12.2.12)$$

This is not only finite, but satisfies the condition

$$\bar{u}(p, \sigma')\Gamma_\mu(p, p)u(p, \sigma) = \bar{u}(p, \sigma')\gamma_\mu u(p, \sigma), \quad (12.2.13)$$

as can also be seen from Eqs. (10.6.13) and (10.6.14). Finally, $\mathcal{L}_2$ makes a contribution $-(Z_3 - 1)(q^2\eta_{\mu\nu} - q_\mu q_\nu)$ to $\Pi_{\mu\nu}^*(q)$. In order that the photon propagator should have a pole with the same residue as for free fields we need the coefficient of $q^2\eta_{\mu\nu} - q_\mu q_\nu$ in the total $\Pi_{\mu\nu}(q)$ to vanish, so

$$Z_3 = 1 + C \quad (12.2.14)$$

and the photon propagator is then finite:

$$\Pi_{\mu\nu}(q) = (\eta_{\mu\nu}q^2 - q_\mu q_\nu)\pi(q^2). \quad (12.2.15)$$

So far, we have only checked that the divergences, arising from the region of momentum space in which all internal momenta are large (and with generic ratios), are polynomials in external momenta that are cancelled by suitable counterterms. Such graphs are called *superficially convergent*. Before we conclude that all ultraviolet divergences actually are removed by renormalization, we need to consider the ultraviolet divergences arising in higher-order graphs when some subset of the momentum space integration variables rather than all of them go to infinity. For instance, in quantum electrodynamics the superficial divergences in subintegrations come from subgraphs that are either photon self-energy parts Π^* , or electron self-energy parts Σ^* , or electron–electron–photon vertices Γ^μ .

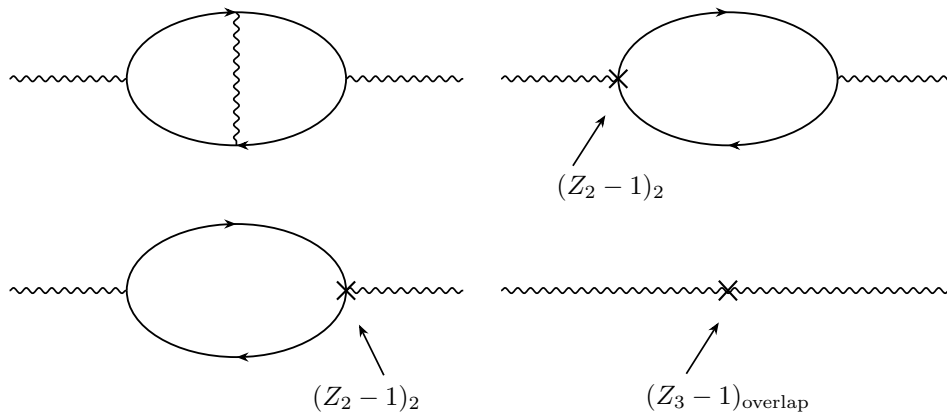


Figure 12.2. Some fourth-order graphs for the photon self-energy in quantum electrodynamics that involve overlapping divergences. Lines carrying arrows are electrons; wavy lines photons. The crosses mark the contribution of counterterms.

The problem with such divergences is that they cannot be removed by differentiation with respect to external momenta; we are left with terms where the derivatives act only on internal lines in the parts of the graph which are *not* in the divergent subgraphs, and therefore do not reduce the degree of divergence of these subgraphs. As mentioned in the previous section, a graph or sum of graphs is actually convergent only if it and all its subintegrations are superficially convergent, in the sense of counting powers of momentum. But wherever such a divergent subgraph appears, it always comes accompanied with an infinite counterterm. In electrodynamics, these are the terms in Eq. (11.1.9): a term $-(Z_3 - 1)(q^2 \eta_{\mu\nu} - q_\mu q_\nu)$ for each $\Pi_{\mu\nu}^*(q)$; a term $Z_2 \delta m - (Z_2 - 1)(i\not{p} + m)$ for each $\Sigma^*(p)$; and a term $(Z_2 - 1)\gamma^\mu$ for each Γ^μ . Just as for the graph as a whole, these counterterms cancel the infinities from the divergent subgraphs.

Unfortunately, there is a flaw in this simple argument—the possibility of overlapping divergences. That is, it is possible that two divergent subgraphs may share an internal line, so that we cannot regard them as independent divergent integrals. In quantum electrodynamics this happens only when two electron–electron–photon vertices overlap inside a photon or an electron self-energy insertion,³ as shown in Figures 12.2 and 12.3.

A complete treatment of renormalization that takes account of overlapping divergences should include a prescription for eliminating superficial ultraviolet divergences, not only in the overall integration but in all subintegrations as well, together with a proof that this prescription is (at least formally) implemented by renormalization of masses, fields, and coupling constants. The theorem of Ref. [2] then ensures that all Green’s functions of renormalized fields are finite when expressed in terms of renormalized masses and couplings.

³The sharing of a line in two self-energy insertions or in a self-energy insertion and a vertex part would not leave enough external lines to attach such a subgraph to the rest of the diagram. Historically, the Ward identity (10.4.26) was used to by-pass the problem of overlapping divergences in the electron self-energy, by expressing the electron self-energy in terms of the vertex function, where overlapping divergences do not occur. This approach will not be followed here, as it is unnecessary, and in any case does not solve the problem for the self-energy of the photon or other neutral particles.

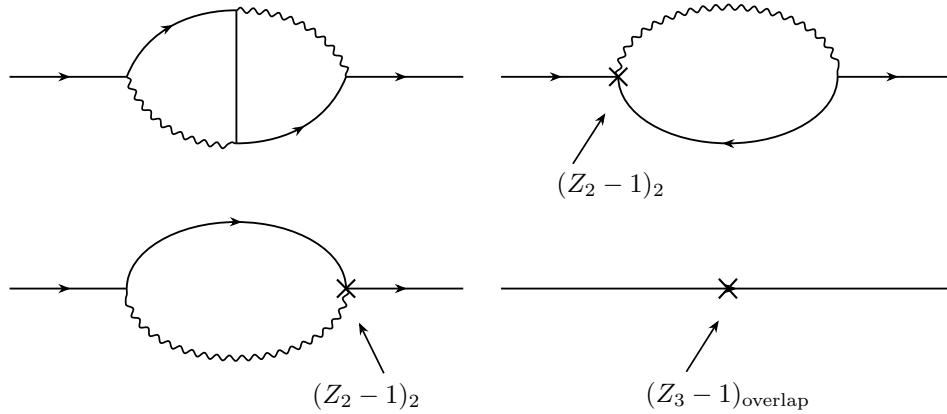


Figure 12.3. Some fourth-order graphs for the electron self-energy in quantum electrodynamics that involve overlapping divergences. Wavy lines are photons, other lines are electrons. The crosses mark the contribution of counterterms.

The first proof that the renormalization of fields, couplings, and masses renders the whole integration and all its subintegrations superficially divergent was offered by Salam [5]. A more specific prescription for eliminating ultraviolet divergences was given by Bogoliubov and Parasiuk [6], and corrected by Hepp [7], and shown by them to be equivalent to a renormalization of fields, masses, and coupling constants. Finally, Zimmermann [8] proved that this prescription does eliminate all superficial divergences in the whole integration and all its subintegrations, and used the theorem of Ref. [2] to conclude from this that the renormalized Feynman momentum-space integrals are convergent.

Briefly, the ‘BPHZ’ prescription for eliminating superficial divergences requires that we consider all possible ways (called ‘forests’) of surrounding a whole graph and/or its subgraphs with boxes that may be nested within each other but do not overlap. (An example is given below.) For each forest we define a subtraction term by replacing the integrand for any subgraph of superficial divergence D within a box (starting with the innermost boxes and working outwards) with the first $D + 1$ terms of its Taylor series expansion in the momenta flowing into or out of that box.⁴ The subtracted Feynman amplitude is given by the original graph, minus all these subtraction terms, including the subtraction term for a forest consisting of a single box surrounding the whole graph.

It is fairly easy to see that the subtracted Feynman amplitude that is calculated in this way is the same as would be obtained by replacing all fields, coupling constants, and masses in the original Lagrangian by their renormalized counterparts. The difference between this procedure and the sort of renormalization we carried out in Chapter 11 is that the renormalized fields, coupling constants, and masses are defined in terms of amplitudes at a unconventional renormalization point, where all four-momenta vanish. (In this respect, the one-dimensional divergent integrals discussed at the start of this section provide an ele-

⁴As described here, this prescription applies to unrenormalizable as well as renormalizable theories. In renormalizable theories it implies that there is no subtraction unless the box contains one of the limited number of graphs corresponding to renormalizable terms in the Lagrangian.

mentary example of the BPHZ method of separating divergent terms.) But there is nothing special about this renormalization point; once a Feynman amplitude is made convergent by expressing it in terms of these unconventional renormalized quantities, it can be rewritten in terms of conventionally renormalized fields, couplings, and masses without introducing new infinities.

It is not necessary to use the BPHZ subtraction prescription in practice. Replacing fields, masses, and couplings with their renormalized counterparts (defined using any convenient renormalization points) automatically provides counterterms that cancel all infinities. Instead of proving that the BPHZ subtraction prescription really does make all integrals converge, we shall just look at one example that shows how renormalization works, even in the presence of overlapping divergences.

Consider the fourth-order contribution to the photon self-energy insertion $\Pi_{\mu\nu}^*(q)$ shown in Figure 12.2. (The forests here consist of the whole integral over p and p' , the subintegration over p alone, and the subintegration over p' alone.) Including the corresponding counterterms for the vertex parts and photon field renormalization, this has the value

$$\begin{aligned} [\Pi_{\mu\nu}^*(q)]_{\text{overlap}} = & -\frac{e^4}{(2\pi)^8} \int d^4p \int d^4p' \frac{1}{(p-p')^2 - i\epsilon} \\ & \cdot \text{Tr}\{S(p')\gamma_\nu S(p'+q)\gamma^\rho S(p+q)\gamma_\mu S(p)\gamma_\rho\} \\ & - 2(Z_2 - 1)_2 \frac{ie^2}{(2\pi)^4} \int d^4p \text{Tr}\{\gamma_\nu S(p+q)\gamma_\mu S(p)\} \\ & - (Z_3 - 1)_{\text{overlap}}(q^2\eta_{\mu\nu} - q_\mu q_\nu), \end{aligned} \quad (12.2.16)$$

where $S(p) \equiv [-i\not{p} + m]/[p^2 + m^2 - i\epsilon]$; $(Z_2 - 1)_2$ is the term in $Z_2 - 1$ of second order in e ; and $(Z_3 - 1)_{\text{overlap}}$ is a logarithmically divergent constant of fourth order in e that cancels the terms in $[\Pi_{\mu\nu}^*(q)]_{\text{overlap}}$ of second order in q^μ . The factor 2 in the second term arises because there is a renormalization counterterm $Z_2 - 1$ for each of the two vertices in the second-order photon self-energy. Note, however, that the first term here can be thought of *either* as the insertion of a vertex correction given by the p' -integral into a photon self-energy given by the p -integral, or as the insertion of a vertex correction given by the p -integral into a photon self-energy given by the p' -integral, but *not* as the insertion of two independent vertex corrections, because there is only one photon propagator.

To see how to handle the infinities in Eq. (12.2.16), note that

$$[(Z_2 - 1)_2 + R_2]\gamma_\mu = \frac{ie^2}{(2\pi)^4} \int \frac{d^4p'}{p'^2 - i\epsilon} \gamma_\rho S(p')\gamma_\mu S(p')\gamma^\rho, \quad (12.2.17)$$

where R_2 is a finite remainder. (Lorentz invariance tells us that the integral on the right is proportional to γ_μ . The difference between this integral and $(Z_2 - 1)_2\gamma_\mu$ equals the complete renormalized electron–electron–photon vertex to second order in e at zero electron and photon momenta, and is therefore finite.) This allows us to rewrite Eq. (12.2.16) in

the form

$$\begin{aligned}
[\Pi_{\mu\nu}^*(q)]_{\text{overlap}} = & -\frac{e^4}{(2\pi)^8} \int d^4p \int d^4p' \left[\frac{1}{(p-p')^2 - i\epsilon} \text{Tr}\{S(p')\gamma_\nu S(p'+q)\gamma^\rho S(p+q)\gamma_\mu S(p)\gamma_\rho\} \right. \\
& - \frac{1}{p'^2 - i\epsilon} \text{Tr}\{S(p')\gamma_\nu S(p')\gamma^\rho S(p+q)\gamma_\mu S(p)\gamma_\rho\} \\
& \left. - \frac{1}{p^2 - i\epsilon} \text{Tr}\{S(p')\gamma_\nu S(p'+q)\gamma^\rho S(p)\gamma_\mu S(p)\gamma_\rho\} \right] \\
& - 2R_2 \frac{ie^2}{(2\pi)^4} \int d^4p \text{Tr}\{\gamma_\nu S(p+q)\gamma_\mu S(p)\} \\
& - (Z_3 - 1)_{\text{overlap}}(q^2\eta_{\mu\nu} - q_\mu q_\nu). \tag{12.2.18}
\end{aligned}$$

First consider integration over p' alone. Each of the first two terms is logarithmically divergent, but their difference is finite. The third term is also logarithmically divergent (with a gauge-invariant regulator), but the divergence in this term (unlike the first two terms) takes the form of a second-order polynomial in q , with the remainder finite. This remaining divergence is cancelled by the term $-(Z_3 - 1)(q^2\eta^{\mu\nu} - q^\mu q^\nu)$ that cancels all second-order terms in the expansion of $\Pi_{\mu\nu}^*(q)$. So the p' -subintegration gives a finite result. The symmetry of Eq. (12.2.16) shows that in exactly the same way the p subintegration also gives a finite result. Generic subintegrations over p and p' with $ap + bp'$ held fixed (where a and b are arbitrary non-zero constants) are manifestly convergent, and the integration over p and p' together is made finite by the counterterm $-(Z_3 - 1)(q^2\eta^{\mu\nu} - q^\mu q^\nu)$. Thus Eq. (12.2.18) and any of its subintegrations satisfy the power-counting requirements for convergence, and therefore according to the theorem [2] quoted in the previous section, the whole expression actually converges.

* * *

In electrodynamics there is a natural definition of renormalized couplings as well as renormalized masses and fields. This is not always the case. For example, consider the theory of a single real scalar field $\phi(x)$ with Lagrangian density

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{1}{24}g\phi^4. \tag{12.2.19}$$

To one-loop order, the S-matrix for scalar-scalar scattering is given by the Feynman rules as

$$S(q_1q_2 \rightarrow q'_1q'_2) = \frac{-i(2\pi)^4\delta^4(p'_1 + p'_2 - p_1 - p_2)}{(2\pi)^6(16E'_1E'_2E_1E_2)^{1/2}} F(q_1q_2 \rightarrow q'_1q'_2), \tag{12.2.20}$$

where

$$\begin{aligned}
-i(2\pi)^4 F(q_1 q_2 \rightarrow q'_1 q'_2) = & -i(2\pi)^4 g + \frac{1}{2} [-i(2\pi)^4 g]^2 \left[\frac{-i}{(2\pi)^4} \right]^2 \\
& \cdot \int d^4 k \left[\frac{1}{[(q_1 + k)^2 + m^2 - i\epsilon][(q_2 - k)^2 + m^2 - i\epsilon]} \right. \\
& \left. + (q_2 \rightarrow -q'_1) + (q_2 \rightarrow -q'_2) \right], \tag{12.2.21}
\end{aligned}$$

and q_1, q_2 and q'_1, q'_2 are the incoming and outgoing four-momenta. Combining denominators and rotating the k^0 -integration contour as usual, this is

$$\begin{aligned}
F = g - \frac{g^2}{16\pi^2} \int_0^\infty k^3 dk \int_0^1 dx \Big\{ [k^2 + m^2 - sx(1-x)]^{-2} \\
+ [k^2 + m^2 - tx(1-x)]^{-2} + [k^2 + m^2 - ux(1-x)]^{-2} \Big\}, \tag{12.2.22}
\end{aligned}$$

where s, t , and u are the Mandelstam variables

$$s = -(p_1 + p_2)^2, \quad t = -(p_1 - p'_1)^2, \quad u = -(p_1 - p'_2)^2, \tag{12.2.23}$$

related by $s + t + u = 4m^2$; also, x is the Feynman parameter introduced in combining denominators. With an ultraviolet cutoff at $k = \Lambda$, this gives the result (for $\Lambda \gg m$)

$$\begin{aligned}
F = g - \frac{g^2}{32\pi^2} \int_0^1 dx \Big\{ \ln \left(\frac{\Lambda^2}{m^2 - sx(1-x)} \right) + \ln \left(\frac{\Lambda^2}{m^2 - tx(1-x)} \right) \\
+ \ln \left(\frac{\Lambda^2}{m^2 - ux(1-x)} \right) - 3 \Big\}. \tag{12.2.24}
\end{aligned}$$

We can define the renormalized coupling g_R as the value of F at any point s, t, u we like, provided we stay in the region where F is real. For instance, suppose that in order to maintain the symmetry among the scalars, we choose to renormalize at the off-mass-shell point ⁵ $p_1^2 = p_2^2 = p'^2_1 = p'^2_2 = \mu^2$, $s = t = u = -4\mu^2/3$. Defining the renormalized coupling g_R as the value of F at this point, we have

$$g = g_R + \frac{3g_R^2}{32\pi^2} \left[\ln \left(\frac{\Lambda^2}{\mu^2} \right) - 1 - \int_0^1 dx \ln \left(\frac{4x(1-x)}{3} + \frac{m^2}{\mu^2} \right) \right] + \dots \tag{12.2.25}$$

The cutoff dependence then cancels in Eq. (12.2.24) to order g_R^2 , leaving a finite formula for F in terms of g_R :

$$\begin{aligned}
F = g_R - \frac{g_R^2}{32\pi^2} \int_0^1 dx \Big\{ \ln \left(\frac{m^2 + 4x(1-x)\mu^2/3}{m^2 - sx(1-x)} \right) + \ln \left(\frac{m^2 + 4x(1-x)\mu^2/3}{m^2 - tx(1-x)} \right) \\
+ \ln \left(\frac{m^2 + 4x(1-x)\mu^2/3}{m^2 - ux(1-x)} \right) \Big\} + \dots \tag{12.2.26}
\end{aligned}$$

⁵Going back over the derivation of Eq. (12.2.25), one may check that in this derivation we have not used the conditions $p_1^2 = p_2^2 = p'^2_1 = p'^2_2 = -m^2$, so Eq. (12.2.24) is valid whatever we take for the external line masses.

Here μ^2 may be taken to be any real quantity greater than $-3m^2$, in which range g_R is real. The explicit μ -dependence in Eq. (12.2.26) is, of course, cancelled by the μ -dependence of the renormalized coupling. This freedom to change the renormalization prescription (which of course exists also in electrodynamics and other realistic theories) will be of great importance to us when we come to the renormalization group method in Volume II.